Product Requirements Document: AI-Powered MSME Financial Health Card

1. Introduction

1.1. Purpose

This Product Requirements Document (PRD) outlines the specifications and requirements for the development of an AI-Powered MSME Financial Health Card. The primary goal is to address the challenges faced by New-to-Credit (NTC) and New-to-Bank (NTB) Micro, Small, and Medium Enterprises (MSMEs) in accessing formal credit by leveraging alternative digital data for comprehensive financial health assessment.

1.2. Scope

The scope of this document covers the aggregation of consented alternate digital data, its transformation into risk-relevant features, the generation of an explainable multidimensional score for MSME financial health and creditworthiness, and the presentation of these insights through a dashboard for lenders. It also includes considerations for compliance, auditability, and integration with India's digital public infrastructure.

1.3. Document Audience

This document is intended for product managers, engineering teams, data scientists, risk analysts, compliance officers, and other stakeholders involved in the design, development, and deployment of the AI-Powered MSME Financial Health Card.

2. Product Overview

2.1. Product Name

AI-Powered MSME Financial Health Card for New-to-Credit and New-to-Bank Enterprises

2.2. Background

Traditional MSME loan applications heavily rely on conventional documents such as audited financial statements, income tax returns, bank statements, collateral records, and bureau history. This approach often excludes a significant number of viable NTC and NTB MSMEs that, despite active operations, lack formal financial depth or prior borrowing records. However, these businesses often generate substantial digital footprints through

GST filings, e-invoices, UPI collections, consented bank transaction data, payroll contributions (EPFO), and utility usage. These digital traces can serve as reliable indicators of cash flow health, operational discipline, business continuity, and repayment capacity. The Account Aggregator framework further facilitates secure, consent-based sharing of such financial information, aligning with digital lending regulations applicable to MSME loans.

2.3. Problem Statement

There is a critical absence of a unified, explainable, and near real-time framework capable of converting diverse alternate digital business data into a lender-ready MSME financial health assessment. This gap leads to several issues for lenders, including slow underwriting processes, high rates of false rejections, limited portfolio diversification, inconsistent risk assessment for thin-file borrowers, and missed opportunities for fostering formal credit inclusion.

2.4. Product Objective

To design and build an AI/ML-powered Financial Health Card that aggregates consented alternate digital data from multiple sources, transforms it into risk-relevant features, and produces an explainable multidimensional score for MSME financial health and creditworthiness. The system aims to support the underwriting and monitoring of businesses with limited traditional credit history, ensuring transparency, auditability, and compliance with consent-led digital lending practices.

2.5. One-line Summary

Build an AI-powered, explainable MSME Financial Health Card that uses consented alternate digital data such as GST, UPI, AA, EPFO, utilities, and bank activity to assess the creditworthiness of New-to-Credit and New-to-Bank businesses in near real time, enabling faster, fairer, and more inclusive lending.

2.6. Polished Problem Statement

Traditional MSME credit appraisal depends heavily on audited statements, collateral, tax filings, and bureau history, which excludes many viable NTC and NTB enterprises from formal credit. Meanwhile, these businesses increasingly leave behind rich digital evidence of commercial activity through GST, digital payments, bank transactions, payroll contributions, and utility consumption. The challenge is to transform this fragmented, alternate data into a unified, explainable, and lender-ready financial health assessment that works in near real time. The proposed solution should aggregate consented data from multiple sources, engineer meaningful business-risk signals, generate a multidimensional Financial Health Score, classify applicants by risk, and present transparent decision-support insights to lenders through a dashboard. The system should be compatible with India's evolving digital lending ecosystem, including Account Aggregator and adjacent infrastructure such as OCEN and ULI, while improving credit access, underwriting speed, portfolio quality, and financial inclusion.

3. Detailed Solution Scope

3.1. Data Sources

The proposed system will support the ingestion of the following data sources, contingent upon user consent and lawful access:

- GST returns and e-invoice-linked business activity.
- UPI collection patterns and digital payment behavior.
- Account Aggregator-based bank transaction data and statement equivalents.
- EPFO contribution trends as a proxy for payroll stability and formal workforce continuity.
- Electricity or utility consumption as an operational activity signal.
- Optional supplementary records such as bank statements, invoice ledgers, Udyam profile, and bureau data, when available.

3.2. Functional Capabilities

The solution will provide the following functional capabilities:

- Aggregate multi-source business data into a common data model.
- Clean, normalize, reconcile, and timestamp data for underwriting use.
- Generate feature sets related to cash flow, compliance, business momentum, volatility, seasonality, resilience, and concentration risk.
- Produce an Overall Financial Health Score along with component sub-scores.
- Classify applicants into low, medium, and high-risk bands.
- Surface reasons behind the score using explainable AI (XAI).
- Present underwriting insights in a comprehensive dashboard for bankers and credit managers.
- Integrate with India's digital public infrastructure direction through Account Aggregator (AA), and remain compatible with Open Credit Enablement Network (OCEN) and Unified Ledger Interface (ULI)-oriented workflows.

3.3. Sample Scoring Dimensions

To avoid a single opaque score, the solution will include weighted dimensions such as:

- Cash-flow strength.
- Revenue consistency.
- Compliance behavior.
- Operational continuity.
- Financial resilience.
- Credit augmentation, where bureau or repayment data exists.

3.4. Expected Outputs

For each MSME, the system is expected to generate:

- Overall Financial Health Score (e.g., 0 to 1000).
- Risk category: Low / Medium / High.
- Top positive drivers for the score.
- Top negative drivers for the score.
- Confidence or data sufficiency indicator.
- Recommended action: approve, approve with conditions, review manually, request more data, or decline.

4. Architecture and Workflow

4.1. Proposed Workflow

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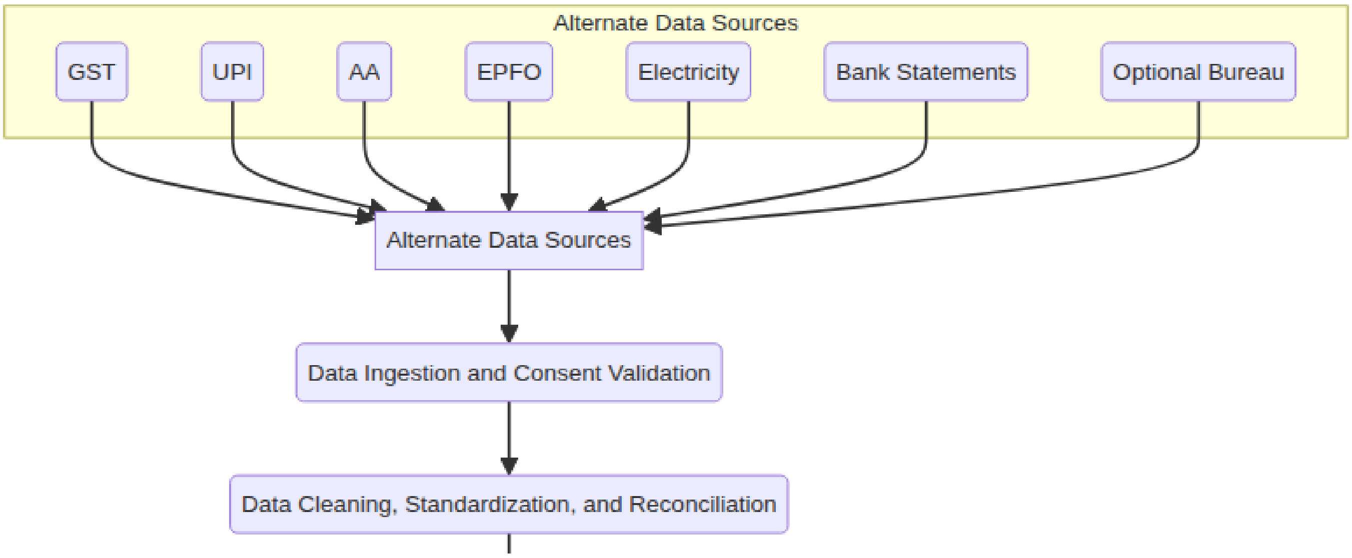

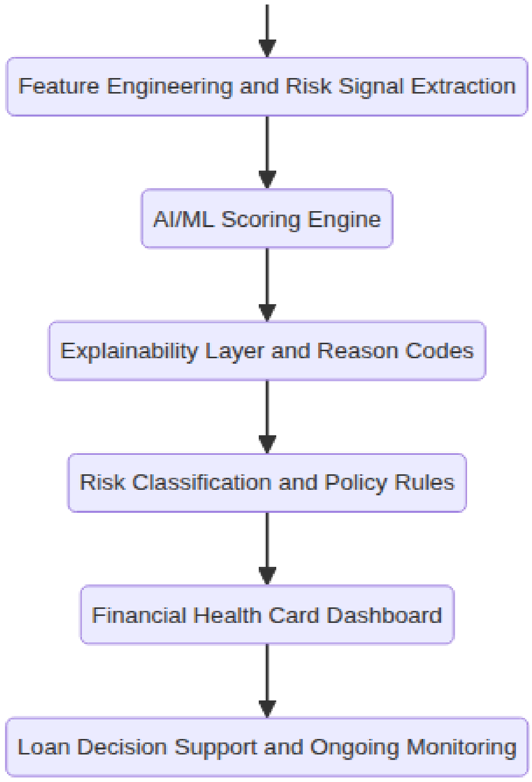

4.2. Suggested Modules

- **Consent Manager:** Captures source-wise, purpose-bound consent and access duration, aligning with AA's explicit, revocable consent model.
- **Source Connectors:** APIs, file uploads, OCR/parsers, AA flows, and billing software adapters for data ingestion.
- **Feature Store:** Maintains reusable and versioned underwriting features.
- **Model Layer:** Combines interpretable scorecards with stronger ML models.
- **Rules Engine:** Overlays policy checks, sector-specific limits, and sanction logic.
- **Audit and Compliance Layer:** Logs consent, model version, decision reasons, and data lineage, emphasizing direct accountability and customer-facing clarity as per RBI's digital lending approach.

5. Key Innovations and Differentiators

- **Near Real-time Refresh:** Scores update as new GST, bank, UPI, or utility signals arrive.
- **Thin-file Underwriting:** Explicitly targets borrowers with little or no bureau depth.
- **Cash-flow Lending Orientation:** Shifts from collateral-heavy lending towards business-activity-based underwriting, leveraging AA-enabled data.
- **Cross-validation Engine:** Compares declared turnover against GST and bank inflows to detect inconsistencies.
- **Fraud and Anomaly Detection:** Identifies sudden invoice spikes, mismatched seasonality, suspicious payer concentration, repeated low-balance stress, and tampered documents.
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- **Early-warning Monitoring:** Utilizes the same framework post-disbursement for monthly deterioration alerts.
- **Human-in-the-loop Underwriting:** AI supports credit officers rather than silently replacing them.
- **Explainability-first Design:** Ensures every adverse or positive inference is traceable to interpretable signals.

6. Evaluation Metrics

A well-written statement should mention measurable success:

- Higher approval rate for creditworthy NTC/NTB MSMEs.
- Lower default or delinquency rate versus baseline underwriting.
- Faster time-to-decision.
- Better precision and recall on risk prediction.
- Higher onboarding of under-documented MSMEs.
- Reduced manual document handling and turnaround time.

7. Technical Considerations

7.1. Suggested Tech Stack

- **AI/ML:** XGBoost, LightGBM, logistic scorecards, anomaly detection.
- **Data Engineering:** Python, Spark or pandas, feature store, API gateway.
- **XAI:** SHAP, reason codes, monotonic constraints, score decomposition.
- **Backend:** FastAPI or Spring Boot.
- **Frontend:** React dashboard with drill-down views.
- **Cloud:** AWS, Azure, or GCP with encryption and audit logging.

8. Anti-Patterns and Considerations

8.1. Product Mistakes to Avoid

- Do not build only a flashy dashboard without a real scoring methodology.
- Do not rely on one data source alone, such as just GST or just UPI.
- Do not produce a black-box score with no reasons, as underwriting requires traceability and defensibility.
- Do not assume missing traditional data always means high risk.
- Do not mix consumer-style behavioral gimmicks with business underwriting unless they are clearly lawful and relevant.

8.2. Compliance and Trust Mistakes to Avoid

- Do not collect more data than necessary for the stated underwriting purpose. RBI digital lending expectations emphasize need-based collection with borrower control and transparency.
- Do not take or store GST portal or bank login credentials as a shortcut.
- Do not scrape portals unofficially when consented or regulated access is expected.
- Do not access phone contacts, call logs, messages, or unrelated device data for lending decisions; such practices are explicitly curbed by digital lending regulations.
- Do not share applicant data with third parties without explicit consent, except where legally required.
- Do not hide decision logic from the lender or borrower-facing process.

8.3. Risk Model Mistakes to Avoid

- Do not train a model without bias checks, calibration checks, and out-of-time validation.
- Do not equate correlation with causation; some alternate signals are noisy proxies.
- Do not ignore seasonality, sector effects, and regional business cycles.
- Do not overfit to synthetic or limited hackathon data and claim production readiness.
- Do not use only approval prediction; include repayment risk or delinquency orientation.

9. Conclusion

The AI-Powered MSME Financial Health Card represents a significant step towards inclusive finance for NTC and NTB enterprises. By leveraging advanced AI/ML techniques and alternative data sources, this product aims to provide a transparent, auditable, and efficient credit assessment solution, ultimately fostering financial inclusion and supporting the growth of the MSME sector. This document serves as a foundational guide for the development team to build a robust and compliant product that meets the stated objectives and addresses the core problem effectively.

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